

Pathology Final Examination

First Lecture: Inflammation

Part I: Multiple Choice Questions (MCQs)

1. Which of the following is considered a non-living chemical irritant that can cause inflammation?

- A) Bacteria and their toxins
 - B) Viruses and fungi
 - ✓ C) Concentrated acids and poisons
 - D) Parasites
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2. Which cell type is mainly responsible for phagocytosis and forms the majority of pus cells?

- A) Lymphocytes
 - ✓ B) Neutrophils
 - C) Plasma cells
 - D) Eosinophils
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3. In chronic inflammation, the cellular infiltrate mainly consists of:

- A) Neutrophils and mast cells
 - ✓ B) Lymphocytes and plasma cells
 - C) Eosinophils only
 - D) RBCs and platelets
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4. All of the following are cardinal signs of inflammation EXCEPT:

- A) Pain
 - B) Heat and redness
 - ✓ C) Pallor**
 - D) Swelling and loss of function
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5. The correct sequence of vascular events in acute inflammation is:

- A) Stasis → Vasodilatation → Vasoconstriction
 - ✓ B) Vasodilatation → Increased vascular permeability → Stasis**
 - C) Exudation → Vasodilatation → Chemotaxis
 - D) Adhesion → Stasis → Vasodilatation
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6. The correct order of neutrophil extravasation is:

- A) Rolling → Adhesion → Margination → Transmigration
 - ✓ B) Margination → Rolling → Adhesion → Transmigration (Diapedesis)**
 - C) Diapedesis → Chemotaxis → Rolling → Adhesion
 - D) Adhesion → Margination → Diapedesis → Rolling
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7. Chemotaxis means:

- A) Ingestion of bacteria
 - ✓ B) Directed movement of inflammatory cells toward the irritant**
 - C) Formation of endothelial gaps
 - D) Fibrin formation
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8. The ingestion and destruction of bacteria and necrotic tissue is called:

- A) Opsonization

B) Diapedesis

✓ C) Phagocytosis

D) Chemotaxis

9. During phagocytosis, wrapping of the cell membrane around the particle is called:

✓ A) Engulfment

B) Recognition

C) Killing

D) Degradation

10. Acute non-suppurative inflammation affecting mucous membranes with excess mucus (e.g. rhinitis) is:

A) Serous inflammation

✓ B) Catarrhal inflammation

C) Allergic inflammation

D) Haemorrhagic inflammation

Part II: Comparison & Short Essay Questions

Q1. Compare between Acute and Chronic Inflammation

Feature	Acute Inflammation	Chronic Inflammation
Onset	Rapid (minutes–hours)	Slow (days)
Cells	Neutrophils	Macrophages & lymphocytes
Tissue injury	Mild, self-limited	Severe, progressive

Feature	Acute Inflammation	Chronic Inflammation
Signs	Prominent	Less prominent

Q2. Define Suppurative Inflammation

Answer:

Suppurative inflammation is an **acute inflammation characterized by pus formation**. It may be **localized** (abscess) or **diffuse** (cellulitis).

Q3. Define Abscess with cause and mechanism

Answer:

An **abscess** is a **localized acute suppurative inflammation** forming a cavity filled with **pus**.

- **Causative organism:** *Staphylococcus*
 - **Mechanism:** Production of **coagulase enzyme** → excess fibrin → localization of infection
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Q4. Define Boil (Furuncle)

Answer:

A boil is a **small abscess** arising from a **hair follicle or sebaceous gland**.

Q5. Compare between Abscess and Cellulitis

Feature	Abscess	Cellulitis
Definition	Localized suppurative inflammation	Diffuse suppurative inflammation
Organism	Staphylococcus	Streptococcus
Mechanism	Excess fibrin (coagulase)	Fibrinolysin & hyaluronidase
Pus	Thick yellow	Thin sanguinous
Spread	Limited	Common